

Causal Inference Assignment 1

Due noon Friday, June 19, 2026

ICPSR Session 1 (June 16, 2026)

1. Refer to the **acorn** data set that accompanies this assignment. Using the mean of turnout proportions in treatment-group precincts, $n_1^{-1} \mathbf{Z}^\top \mathbf{y}$, as test statistic, simulate its randomization distribution under the sharp null hypothesis of no effect, reporting:

- (a) your simulation p -value;
- (b) your simulation approximation of $E[n_1^{-1} \mathbf{Z}^\top \mathbf{y}]$, the null expected value of the test statistic;
- (c) your simulation approximation of $\text{Var}[n_1^{-1} \mathbf{Z}^\top \mathbf{y}]$, this test statistic's variance under the null.

Here n_1 is the number of treated units, $n_0 = N - n_1$ the number of control units, N the total number of units, $\mathbf{Z}$ the random assignment vector, and $\mathbf{y}$ the observed outcome vector. In the **acorn03** data set, the column **z** records the observed treatment assignment z (the realized GOTV assignment indicator), of which $\mathbf{Z}$ is the random counterpart that you simulate over; the column **vote03** records the observed outcome y (the 2003 turnout proportion in each precinct).

2. Calculate $E[n_1^{-1} \mathbf{Z}^\top \mathbf{y}]$, the expected value of the treatment-group mean under the sharp null hypothesis of no effect, from first principles — i.e., without simulations — using the data in $\mathbf{y}$.
3. In this setup, $\text{Var}[\bar{y}_1] = \frac{1}{n_1} \frac{n_0}{N} \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N-1}$, where $\bar{y}_1 = n_1^{-1} \mathbf{Z}^\top \mathbf{y}$ is the mean of y s among the group assigned to treatment ($\{i : Z_i = 1\}$) while $\bar{y} = N^{-1} \mathbf{1}^\top \mathbf{y}$ is the mean of y over the full study population. If you've seen formulas for the sampling variance of the mean in earlier stats courses, you probably saw

$$\widehat{\text{Var}}[\bar{y}_1] = \frac{s_y^2}{n_1} = \frac{1}{n_1} \frac{\sum_{i=1}^{n_1} (y_i - \bar{y}_1)^2}{n_1 - 1}, \text{ and/or } \text{Var}[\bar{y}_1] = \frac{\sigma_y^2}{n_1} = \frac{1}{n_1} \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N}.$$

Which gives a larger value, our new $\text{Var}[\bar{y}_1]$ -formula or the $\text{Var}[\bar{y}_1]$ formula immediately above? Explain why it makes sense that the formula we've given would differ in the direction it does from this other formula.

4. Calculate $\text{Var}[n_1^{-1} \mathbf{Z}^\top \mathbf{y}]$ using the appropriate formula. Determine the error of the simulation-based approximation to this quantity that you reported in question 1, expressing it as a percentage of $\text{Var}[n_1^{-1} \mathbf{Z}^\top \mathbf{y}]$.
5. Determine the Normal-theory approximation to $\Pr(n_1^{-1} \mathbf{Z}^\top \mathbf{y} \geq n_1^{-1} \mathbf{z}^\top \mathbf{y})$. (Hints: Use the variance and expected value calculated in 4 and 2 to transform your observed treatment-group mean into a corresponding “z-score.” To determine Normal quantiles corresponding to z-scores in R, use `pnorm()`; type `?pnorm` for help. Here $\mathbf{z}^\top$ refers to the observed ACORN treatment assignment.)
6. A researcher plans to ask six subjects to donate time to an adult literacy program. Each subject will be asked to donate either 30 ($Z = 0$) or 60 ($Z = 1$) minutes. The researcher is considering three methods for randomizing the treatment. Method I is to make independent decisions for each subject, tossing a coin each time. Method C is to write “30” and “60” on three playing cards each, and then shuffle the six cards. Method P tosses one coin for each of the 3 pairs (1, 2), (3, 4), (5, 6), asking for 30 (60) minutes from exactly one member of each pair.

- a Discuss strengths & weaknesses of each method.
- b How would your answers to (a) change if $N : 6 \mapsto 600$?
- c Determine $E[Z_1]$ under each method.
- d Determine $E[Z_1 + Z_2 + \dots + Z_6]$ under each method.
- e Calculate $E[\mathbf{Z}^\top \mathbf{1}]$ under each of the three methods.
- f For which of the methods does $E[(\mathbf{Z}^\top \mathbf{1} - E[\mathbf{Z}^\top \mathbf{1}])^2] = 0$?¹
- g For two of the three methods, the linearity property of expected value² entails that $E[\frac{\mathbf{Z}^\top \mathbf{x}}{\mathbf{Z}^\top \mathbf{1}}] = (x_1 + x_2 + \dots + x_6)/6$. Which two are these, and why doesn't the argument work for the third?

¹I.e., for which does $\text{Var}[\mathbf{Z}^\top \mathbf{1}] = 0$? (In general, $\text{Var}[V] = E[(V - E[V])^2]$.)

²For constants α and β , $E(\alpha + \beta Y) = \alpha + \beta E(Y)$. Also $E(\alpha X + \beta Y) = \alpha E(X) + \beta E(Y)$.